

HEEJO JEONG

wjdgmlwh1629343@gmail.com [github](#) [homepage](#)

RESEARCH INTERESTS

Computer Graphics, Rendering, Physics-based Animation, Numerical Methods, Optimization

EDUCATION

Korea University

Seoul, Republic of Korea

M.S. in Computer Science and Engineering

Sep. 2023 - Aug. 2025

- Advisor: Prof. [JungHyun Han](#)
- GPA: 4.34 / 4.5

Korea University

Seoul, Republic of Korea

B.S. in Artificial Intelligence (Interdisciplinary Program)

Sep. 2019 - Aug. 2023

B.S. in Civil, Environmental and Architectural Engineering

Mar. 2017 - Aug. 2023

- Including two years of military service
- GPA: 4.06 / 4.5

PUBLICATIONS

- [1] [Momentum-preserving Inversion Alleviation for Elastic Material Simulation](#)

Heejo Jeong, Seung-wook Kim, JaeHyun Lee, Kiwon Um, Min Hyung Kee, and JungHyun Han, Computer Animation and Virtual Worlds (CAVW), Vol. 35, No. 3, May 2024, pp. e2249.

- Poster version presented at the Korea Computer Graphics Society (KCGS), July, 2024.

RESEARCH & PROJECT

M.S. Thesis: Learning Neural Hyperelastic Constraints in XPBD Simulation

Researcher

Jul. 2024 - Feb. 2025

- Implemented a differentiable XPBD solver for learning hyperelastic constraints from a single motion trajectory. [[Korea Univ. Library](#)]

Constraining Velocities in Position-based Dynamics

Research Assistant

Mar. 2024 - May 2024

- Developed a velocity-based extension of the PBD framework to enhance stability and efficiency in dynamic simulations.
- Submitted to SIGGRAPH Asia 2024.

Dimension Expansion for Finite Element Method-based Deformable Body Simulation

Researcher

Sep. 2023 - Dec. 2023

- Extended a finite element method-based deformable body simulation into 4D by introducing auxiliary coordinate and projecting the result back to 3D for rendering.

LG Electronics: Real-time Air Conditioning Airflow Simulation and Visualization

Project Assistant

Jun. 2023 - Sep. 2023

- Implemented a Poisson solver using preconditioned conjugate gradient method with a sparse matrix for fluid simulation.

Real-time Vision-based Human Pose Matching Framework

Research Assistant

Mar. 2023 - May 2023

- Implemented a multiple human tracking module using YOLOv7 and Kalman filter.

EXPERIENCE

Research Associate , Korea Univ., Korea	<i>Sep. 2025 – Dec. 2025</i>
Research Intern , Télécom Paris, Institut Polytechnique de Paris, France	<i>Jan. 2025 – Feb. 2025</i>
Research Intern , Korea Univ., Korea	<i>Sep. 2023 – Aug. 2024</i>

SCHOLARSHIPS

Research Assistant Scholarship, Korea University	<i>Fall 2023 – Spring 2025</i>
Research Scholarship, Korea University	<i>Fall 2023 – Fall 2024</i>
BK21 FOUR Outstanding New Student Scholarship	<i>Fall 2023</i>
National Grant	<i>Spring 2017 – Fall 2019, Fall 2022 – Spring 2023</i>
KU Alumni Scholarship, Korea University	<i>Spring 2023</i>
POSCO Scholarship, POSCO	<i>Fall 2022</i>
Special Scholarship, Korea University	<i>Spring 2019</i>
Study Scholarship, Korea University	<i>Fall 2018, Fall 2019</i>

HONORS

Semester High Honors	<i>Spring 2018, Fall 2018, Fall 2022, Spring 2023</i>
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TECHNICAL SKILLS

Languages: C/C++, Python, MATLAB
APIs: OpenGL, OpenMP, CUDA, Taichi Lang, PyTorch
Software: LaTeX, Blender, MS Office, Photoshop

LANGUAGE LEVEL

Korean: Native
English: Fluent (TOEFL iBT 97)